

ML ACADEMY

Generative AI Interview Handbook

Detailed Questions, Explanations & Worked Examples for Top Tech
Interviews

Generative Models • GANs • VAEs • Diffusion • Multimodal AI Foundations • Training • Sampling •
Evaluation • Safety • Production

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How to get the most from this handbook

This PDF is designed for serious interview preparation. Each topic combines an interview-ready answer with deeper explanation, examples, diagrams, code, calculations or system trade-offs wherever they improve understanding.

Recommended approach: answer the question yourself first, then compare with the explanation and work through the example.

For coding and SQL questions, state assumptions, approach, complexity and edge cases before writing code. For ML/AI system design, discuss requirements, trade-offs, failure modes, monitoring and security instead of claiming one architecture is always best.

Section	Topics
1. Research basis and further reading	1
TOTAL COVERAGE	1

1. Generative modeling foundations

Start by understanding what distribution or transformation the model is learning, then explain how samples are generated.

Interview framing Generative AI questions are strongest when you compare model families by objective, sampling process, controllability, evaluation, training cost and failure modes - not by listing acronyms.

INTERVIEW QUESTION

Generative model vs discriminative model: what is the difference?

Interview-ready answer

A discriminative model learns to predict a target or decision boundary from inputs; a generative model learns enough structure about the data distribution to create new samples or model how observations could have been produced.

Detailed explanation

A classifier may learn $p(y|x)$ or a decision function, while a generative model may model $p(x)$, $p(x,y)$, a latent-variable likelihood, an adversarial generator, or a score/noise-prediction process. The practical distinction is the task: prediction versus synthesizing, reconstructing or sampling plausible data. Modern foundation models can do both through prompting, so describe the training objective and generation mechanism rather than forcing every model into one label.

Example

A spam classifier maps an email to spam/not-spam. A language model generates a continuation token by token. A diffusion image model starts from noise and iteratively denoises toward a sample conditioned on text.

Likely follow-up questions

- Can a generative model be used for classification?
- Why is likelihood not available for every generative family?
- What does sampling mean?

Weak answer to avoid

Saying generative models simply 'create content' and discriminative models 'do not' without discussing the learned objective.

Best interview habit

Answer the main question first, then volunteer one trade-off and pause for the interviewer.

INTERVIEW QUESTION

How does a Variational Autoencoder (VAE) work?

Interview-ready answer

A VAE encodes an input into a probability distribution in a latent space, samples a latent vector using the reparameterization trick, and decodes it back to reconstruct the input. Training balances reconstruction quality with keeping the latent distribution close to a simple prior.

Detailed explanation

The encoder predicts mean μ and scale σ for $q(z|x)$. Instead of sampling z directly in a way that blocks gradients, write $z = \mu + \sigma * \epsilon$ with ϵ sampled from a fixed standard normal. The loss combines a reconstruction term with a KL-divergence regularizer that keeps $q(z|x)$ near $p(z)$, usually $N(0,I)$. This creates a smoother latent space that can be sampled, unlike a plain deterministic autoencoder.

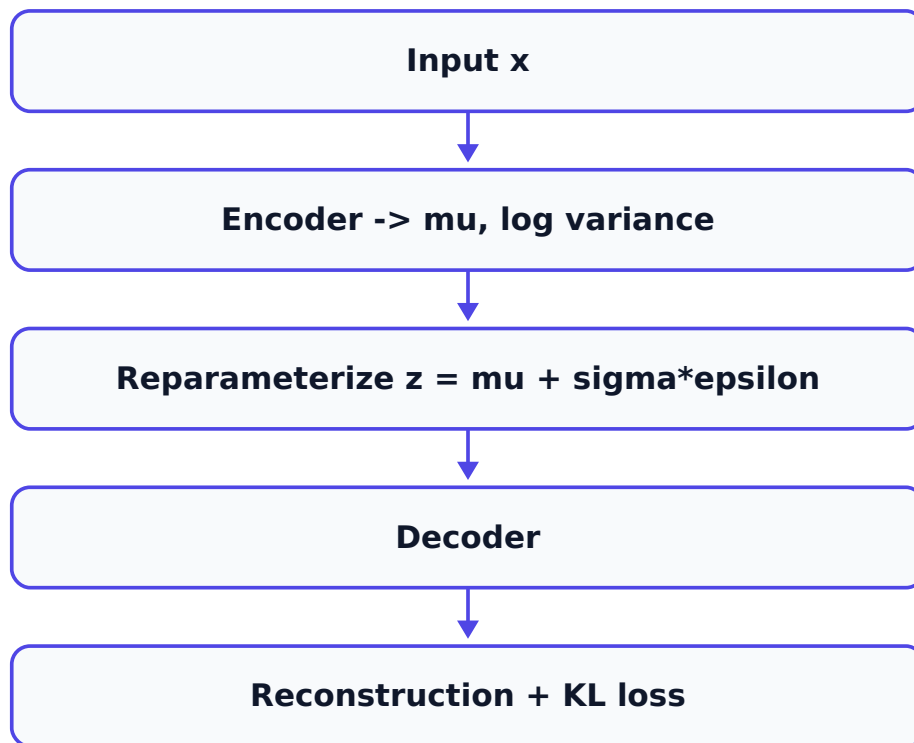


Figure: conceptual flow for this answer.

Example

If the encoder outputs $\mu=[0.4,-0.2]$ and $\sigma=[0.5,0.8]$, sampling ϵ from $N(0,1)$ gives a differentiable z through μ and σ . At generation time, sample z from $N(0,1)$ and decode it.

```
# simplified VAE reparameterization
def reparameterize(mu, logvar):
    std = torch.exp(0.5 * logvar)
    eps = torch.randn_like(std)
    return mu + eps * std

# loss = reconstruction_loss + beta * KL(q(z|x) || N(0,I))
```

Likely follow-up questions

- Why does the KL term help generation?
- What happens if KL dominates too strongly?
- What is posterior collapse?

Weak answer to avoid

Calling a VAE just an autoencoder with random noise; the probabilistic latent objective is central.

Best interview habit

Answer the main question first, then volunteer one trade-off and pause for the interviewer.

INTERVIEW QUESTION

How does a GAN train, and why is mode collapse a problem?

Interview-ready answer

A GAN trains a generator to produce samples that fool a discriminator while the discriminator learns to distinguish real from generated samples. Mode collapse occurs when the generator produces only a narrow subset of the real data distribution.

Detailed explanation

Training is a minimax game. The discriminator provides a learned signal rather than a fixed pixel-level loss, which can produce sharp samples but makes optimization unstable. If the generator discovers a few outputs that consistently fool the discriminator, it may ignore other modes. Better objectives, regularization, architecture, augmentation and training balance can help, but GAN stability remains more delicate than many likelihood- or diffusion-based approaches.

Example

A face GAN trained on diverse people might generate many sharp faces but repeatedly produce similar hairstyles and poses. Visual quality looks high while coverage is poor.

Likely follow-up questions

- Why can the discriminator become too strong?
- How would you detect mode collapse?
- GAN or diffusion model: what trade-offs matter?

Weak answer to avoid

Equating mode collapse with ordinary overfitting; it specifically describes poor coverage of output modes.

Best interview habit

Answer the main question first, then volunteer one trade-off and pause for the interviewer.

2. Diffusion models and controlled generation

Diffusion interviews often move from the forward/reverse process to speed, guidance and conditioning.

INTERVIEW QUESTION2026 SIGNAL

Diffusion remains a core modern GenAI concept and official Diffusers documentation centers training around noise schedules, denoising targets and memory-efficient training.

Explain diffusion models from training to generation.

Interview-ready answer

During training, noise is added to clean data at randomly sampled timesteps and a neural network learns to predict a denoising target. During generation, start from noise and repeatedly apply the learned reverse process until a sample emerges.

Detailed explanation

The forward diffusion process is known and progressively destroys structure. The model, commonly a U-Net or Transformer-like denoiser, receives a noisy sample, timestep and optional condition such as text. Many implementations predict epsilon, velocity, or an equivalent target. At inference, a scheduler converts those predictions into a sequence of denoising updates. Latent diffusion performs this process in a compressed latent space to reduce compute.

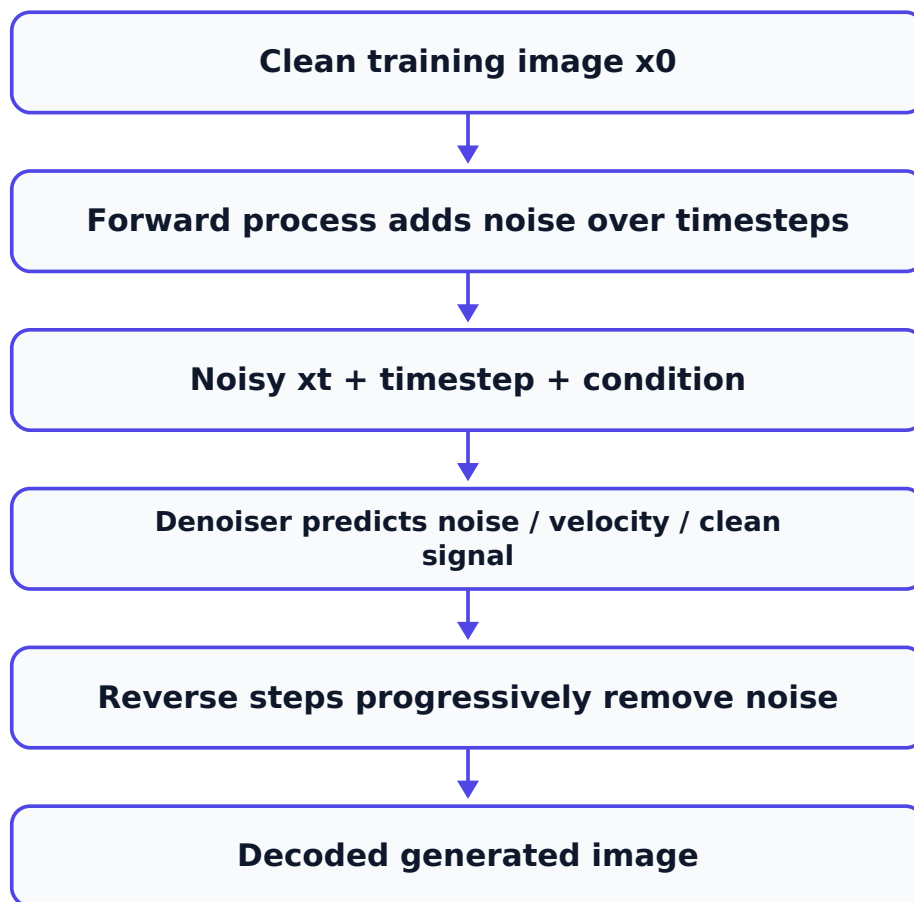


Figure: conceptual flow for this answer.

Example

Training does not require running the full 50-step reverse chain for every example. A random timestep t is sampled, noise is added directly to x_0 according to the schedule, and the network learns the denoising target for that t .

Likely follow-up questions

- Why can diffusion be slow at inference?
- What is latent diffusion?
- What is the role of the scheduler?

Weak answer to avoid

Saying the model memorizes how to 'remove Gaussian noise' without explaining timestep conditioning and the reverse sampling process.

Best interview habit

Answer the main question first, then volunteer one trade-off and pause for the interviewer.

INTERVIEW QUESTION

What is classifier-free guidance (CFG)?

Interview-ready answer

Classifier-free guidance combines conditional and unconditional denoising predictions so generation follows the condition more strongly without needing a separate classifier.

Detailed explanation

A simplified form is $\text{eps_guided} = \text{eps_uncond} + s * (\text{eps_cond} - \text{eps_uncond})$. Guidance scale s controls how strongly the sample is pushed toward the condition. Higher guidance can improve prompt alignment but may reduce diversity, exaggerate artifacts or oversaturate the sample. The best value depends on model and scheduler.

Example

If $s=1$, the result is close to the conditional prediction. Raising s increases the conditional direction. Pushing s too high can make images look less natural even when the prompt match improves.

Likely follow-up questions

- Why train with dropped conditioning?
- What happens to diversity as guidance increases?
- Is the exact CFG formula identical in every parameterization?

Weak answer to avoid

Calling guidance a temperature setting. It changes the denoising direction by contrasting conditional and unconditional predictions.

Best interview habit

Answer the main question first, then volunteer one trade-off and pause for the interviewer.

INTERVIEW QUESTION

Why use latent diffusion instead of pixel-space diffusion?

Interview-ready answer

Latent diffusion runs the denoising process in a learned compressed representation, reducing spatial dimensionality and therefore compute and memory while preserving enough perceptual information for high-quality decoding.

Detailed explanation

An encoder maps the image to latent features, the diffusion model operates there, and a decoder reconstructs pixels. Compression trades some exact pixel detail for much cheaper generative modeling. The autoencoder itself therefore becomes part of the quality pipeline: poor latent reconstruction limits what diffusion can ultimately generate.

Example

Instead of denoising a 1024x1024x3 pixel tensor at every step, a latent model may denoise a much smaller spatial grid with several latent channels, then decode once near the end.

Likely follow-up questions

- Where can artifacts enter the pipeline?
- Why is the VAE decoder important in Stable Diffusion-style systems?

Weak answer to avoid

Saying latent diffusion is just image compression after generation; the diffusion process itself happens in latent space.

Best interview habit

Answer the main question first, then volunteer one trade-off and pause for the interviewer.

3. Fine-tuning, control and evaluation

Modern GenAI products combine a base model with lightweight adaptation, controls and evaluation gates. Interview answers should show that you can choose the smallest intervention that solves the failure.

INTERVIEW QUESTION2026 SIGNAL

Current Hugging Face Diffusers documentation presents LoRA as a lightweight technique that reduces trainable parameters and produces small adapter weights.

LoRA vs full fine-tuning: when would you choose each?

Interview-ready answer

Full fine-tuning updates most or all model weights for maximum flexibility but is expensive. LoRA inserts low-rank trainable adapters into selected layers, dramatically reducing trainable parameters, optimizer memory and artifact size.

Detailed explanation

LoRA approximates an update matrix as a product of two low-rank matrices. Rank controls adapter capacity. It is useful when you need multiple specialized variants of a large model, have limited GPU memory, or want small portable deltas. Full fine-tuning can still win when the target task/domain differs substantially and you have enough high-quality data and compute.

Example

Instead of storing several multi-gigabyte full copies of an image model, a team can store one base model plus small LoRA adapters for product styles, provided evaluation confirms enough quality.

```
# conceptual PEFT setup
from peft import LoraConfig
config = LoraConfig(
    r=16,
    lora_alpha=16,
    target_modules=["to_q", "to_k", "to_v", "to_out.0"]
)
# attach adapter to supported model, train only adapter params
```

Likely follow-up questions

- What does LoRA rank control?
- Why can LoRA use a larger learning rate?
- What are the operational advantages of adapter versioning?

Weak answer to avoid

Saying LoRA is always equivalent to full fine-tuning in quality.

Best interview habit

Answer the main question first, then volunteer one trade-off and pause for the interviewer.

INTERVIEW QUESTION2026 SIGNAL

Official Diffusers documentation describes ControlNet as an adapter trained over a pretrained model using extra conditioning images such as edges, depth and pose.

What problem does ControlNet solve?

Interview-ready answer

ControlNet adds spatial conditioning to a pretrained diffusion model so generation can follow structure such as edges, depth, pose or segmentation while preserving the capabilities of the base model.

Detailed explanation

A text prompt often describes semantics but not precise geometry. ControlNet processes an additional conditioning image and injects its signal into the denoising network through trainable control branches. This allows a system to say not only 'generate a person' but also 'follow this pose or depth layout'.

Example

For architectural rendering, text describes materials and style while a depth map constrains the building geometry. For character generation, a pose map can constrain limb positions.

Likely follow-up questions

- Why not just include an edge map in the text prompt?
- What happens if the control signal conflicts with text?
- How does training memory compare with LoRA?

Weak answer to avoid

Calling ControlNet a post-processing filter; it conditions the denoising process.

Best interview habit

Answer the main question first, then volunteer one trade-off and pause for the interviewer.

INTERVIEW QUESTION

How do you evaluate a generative image model?

Interview-ready answer

Use multiple dimensions: fidelity, diversity, condition/prompt alignment, human preference, safety, latency and cost. No single automatic metric captures all desired behavior.

Detailed explanation

Distributional metrics can compare generated and reference features, similarity models can estimate text-image alignment, and human raters can judge preference or defects. For a product, create a representative prompt set with slices for people, text rendering, rare concepts, styles, safety-sensitive content and hard compositions. Track regression by slice rather than only one global score.

Example

A new model may improve prompt alignment while producing less diverse images or more artifacts in hands and text. A single average preference score can hide the regression.

Likely follow-up questions

- What are weaknesses of FID?
- How would you evaluate memorization or privacy risk?
- How would you compare two models with human raters?

Weak answer to avoid

Using one metric such as CLIP similarity as the definition of overall quality.

Best interview habit

Answer the main question first, then volunteer one trade-off and pause for the interviewer.

INTERVIEW QUESTION

How do you decide between prompting, ControlNet, LoRA and full fine-tuning?

Interview-ready answer

Match the intervention to the failure: improve instructions with prompting; control spatial structure with conditioning such as ControlNet; adapt style or domain behavior cheaply with LoRA; use broader fine-tuning when the base representation or behavior needs deeper change.

Detailed explanation

The engineering decision should include data requirements, training cost, inference cost, number of variants, artifact management and evaluation. Often the best solution combines methods: a strong prompt + a LoRA + structural control.

Example

A retail team wants product images in the same brand style and exact pose. Use a LoRA for brand style and pose/depth ControlNet for geometry rather than full fine-tuning the entire base model for every campaign.

Likely follow-up questions

- When can LoRA overfit?
- What if the product catalog changes weekly?
- How do you version multiple adapters?

Weak answer to avoid

Choosing the most expensive fine-tuning method before diagnosing whether the problem is instruction, structure or domain adaptation.

Best interview habit

Answer the main question first, then volunteer one trade-off and pause for the interviewer.

4. Production Generative AI

At senior levels, the interview usually moves beyond model families into serving, safety, provenance and cost.

INTERVIEW QUESTION

Design a production text-to-image service.

Interview-ready answer

Separate model preparation from online serving. Online, validate the request, enforce policy, route to model/adapters, generate under resource budgets, validate the output, attach provenance metadata and log quality/safety signals.

Detailed explanation

The service needs a request queue because generation is GPU-heavy and variable. Use admission control and quotas to prevent overload. Cache reusable assets carefully, version models/adapters/schedulers together, and preserve enough metadata to reproduce an output. Safety is layered: prompt checks, model behavior, output classifiers, policy rules and human escalation for edge cases.

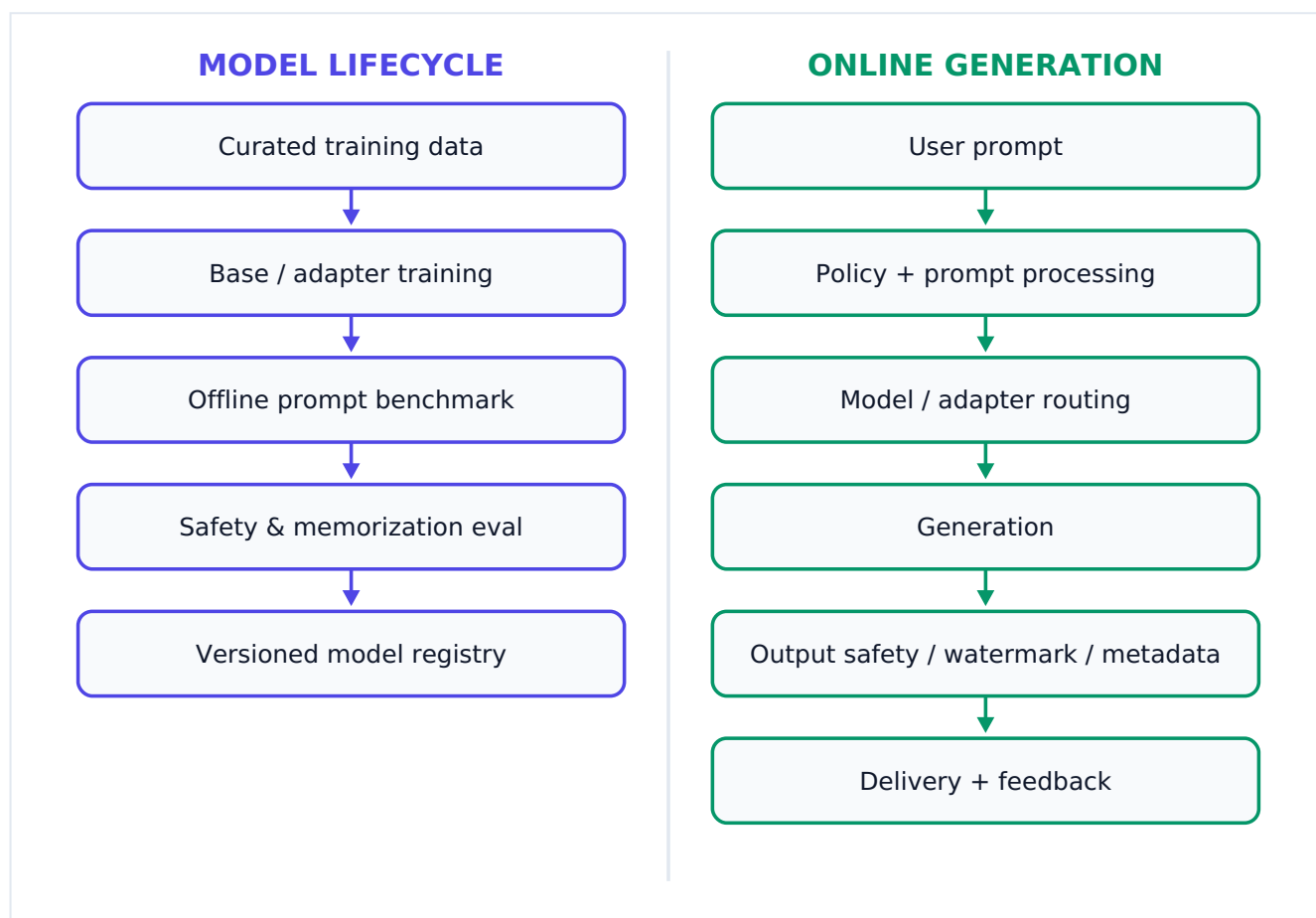


Figure: conceptual flow for this answer.

Example

A marketplace lets sellers create lifestyle images. The service authenticates the seller, checks the prompt, chooses the catalog-specific adapter, allocates a generation job, runs the image model, scans the output, stores model/prompt/version metadata and returns the image URL.

Likely follow-up questions

- How would you set GPU capacity?
- What if safety filtering rejects too many benign images?
- How do you reproduce a customer complaint?

Weak answer to avoid

Drawing only 'prompt -> model -> image' and ignoring queuing, safety, versioning, cost and observability.

Best interview habit

Answer the main question first, then volunteer one trade-off and pause for the interviewer.

INTERVIEW QUESTION

What are the main production failure modes for generative-image systems?

Interview-ready answer

Group failures into content quality, safety/policy, reliability/performance, data/privacy and product economics. Each class needs different evidence and controls.

Detailed explanation

Quality failures include prompt mismatch, composition defects and style drift. Safety includes disallowed content and biased outputs. Reliability includes GPU OOM, queue growth and timeouts. Privacy includes memorization or training-data provenance concerns. Economics includes runaway retries, oversized images and expensive default settings. Monitor by prompt slice, model version and serving path.

Example

If p95 generation latency jumps only for 2048px requests, the fix may be resolution-aware admission control or a different pipeline - not model retraining.

Likely follow-up questions

- How would you detect memorization?
- Which failures can be caught before model execution?
- How would you roll back an adapter?

Weak answer to avoid

Treating every bad output as a model-quality problem.

Best interview habit

Answer the main question first, then volunteer one trade-off and pause for the interviewer.

INTERVIEW QUESTION

Why are reproducibility and provenance difficult in generative systems?

Interview-ready answer

Outputs depend on model version, adapter version, scheduler, random seed, precision, prompt processing and sometimes nondeterministic kernels. Reproducing an output therefore requires capturing the full generation configuration, not just the prompt.

Detailed explanation

For incident response, save identifiers rather than every internal tensor: model/adapter hashes, scheduler and step count, seed, safety-policy version, input asset IDs and relevant environment version. Exact bit-for-bit replay may still be impossible across hardware/software changes, so define the reproducibility level the product actually requires.

Example

A user reports an unsafe image. The audit record should let you reconstruct which base model, LoRA, seed, prompt template and safety policy were used, then rerun a comparable job.

Likely follow-up questions

- Should you log raw user prompts?
- How do privacy rules affect observability?
- Why can deterministic flags reduce performance?

Weak answer to avoid

Assuming the random seed alone is enough to reproduce output.

Best interview habit

Answer the main question first, then volunteer one trade-off and pause for the interviewer.

Deep-dive Generative AI tutorial: VAEs, GANs, diffusion and production

Use this section to understand how major generative model families work, how they are controlled and adapted, and how production systems evaluate and serve them.

Study method Do not memorize definitions only. For each concept, explain the idea, work the example, name a failure mode, and state one trade-off. That pattern matches how stronger technical interviews are evaluated.

1. What makes a model generative?

A generative model learns a data distribution or a mechanism that can sample new examples resembling the training distribution. A discriminative model focuses on predicting labels or decision boundaries. Modern generative systems can model text, images, audio, video or multimodal outputs.

The interview distinction is about the learned objective and sampling process, not whether the final application “creates” something.

2. Autoencoders and latent representations

An autoencoder compresses an input into a latent representation and reconstructs it. A plain autoencoder is not automatically a well-behaved generative model because its latent space may contain gaps with no meaningful samples.

Variational Autoencoders address this by learning distributions for latent variables and regularizing them toward a prior.

3. VAE objective and the reparameterization trick

A VAE balances reconstruction quality with a KL-divergence term that keeps the approximate posterior near a prior such as $N(0, I)$. The reparameterization trick writes $z = \mu + \sigma \odot \epsilon$ with ϵ sampled from a fixed distribution, allowing gradients to flow through μ and σ .

The ELBO can be understood as reconstruction term plus latent regularization; overly strong KL pressure can cause posterior collapse.

Worked example / interview scenario

Encoder outputs $\mu=[0.3, -0.2]$ and log-variance values. Sample ϵ , construct z differentiably, decode z , and backpropagate through μ and σ .

4. GAN training as a two-player game

A generator maps noise to synthetic samples while a discriminator distinguishes real from generated data. The generator improves by making samples the discriminator classifies as real. Training can be unstable because both players change simultaneously.

Mode collapse occurs when the generator covers only a small subset of the data distribution. Remedies include better objectives, regularization, architecture choices and monitoring diversity.

5. Diffusion: forward noising and reverse denoising

Diffusion models define a forward process that gradually adds noise and train a network to predict noise, velocity or another parameterization needed to reverse the process. Generation starts from noise and iteratively denoises.

Noise schedules control how signal-to-noise ratio changes across timesteps. Sampling can use fewer steps than training through specialized solvers, trading speed against fidelity.

Worked example / interview scenario

At high noise, global structure is uncertain. As denoising progresses, coarse composition appears before fine textures. This intuition explains why conditioning can steer the trajectory.

Deep dive continued

6. Classifier-free guidance

Classifier-free guidance combines conditional and unconditional model predictions to strengthen alignment with the prompt. Increasing guidance can improve prompt adherence but may reduce diversity or create artifacts when pushed too far.

It is a controllability knob, not a universal quality slider.

7. Latent diffusion

Pixel-space diffusion is expensive because high-resolution images contain many values. Latent diffusion first encodes the image to a lower-dimensional latent space, runs diffusion there, and decodes back to pixels.

This reduces compute while preserving much of the perceptual information, which made high-resolution diffusion systems more practical.

8. Text conditioning and cross-attention

Text-to-image systems encode the prompt into text representations and use cross-attention so visual latents can attend to prompt tokens. Prompt wording therefore influences which concepts, attributes and relations receive attention.

Long or contradictory prompts can create competition; structured prompting helps but cannot guarantee exact composition.

9. LoRA, adapters and full fine-tuning

LoRA learns low-rank updates to selected weight matrices while keeping most base weights frozen. It reduces trainable parameters and storage, enabling multiple lightweight adaptations. Full fine-tuning changes all weights and may achieve stronger adaptation at much higher compute and forgetting risk.

Choose based on data volume, desired behavioral change, hardware and deployment needs.

10. ControlNet and structural conditioning

ControlNet adds a conditioning pathway so generation can follow structure such as edges, pose, depth or segmentation. It solves a different problem from LoRA: ControlNet controls geometry/layout at inference, while LoRA adapts learned style/concepts or behavior.

An interview answer should distinguish “what to generate” from “how to constrain structure”.

Deep dive continued

11. Evaluation: quality, diversity and alignment

Generative evaluation is multi-dimensional. Automated metrics such as FID or embedding similarity capture limited aspects. Human evaluation may assess realism, prompt adherence, preference, safety and diversity.

A production evaluation suite should include representative prompts, edge cases, identity/safety checks, latency and cost. One score rarely captures user quality.

12. Safety, provenance and production service design

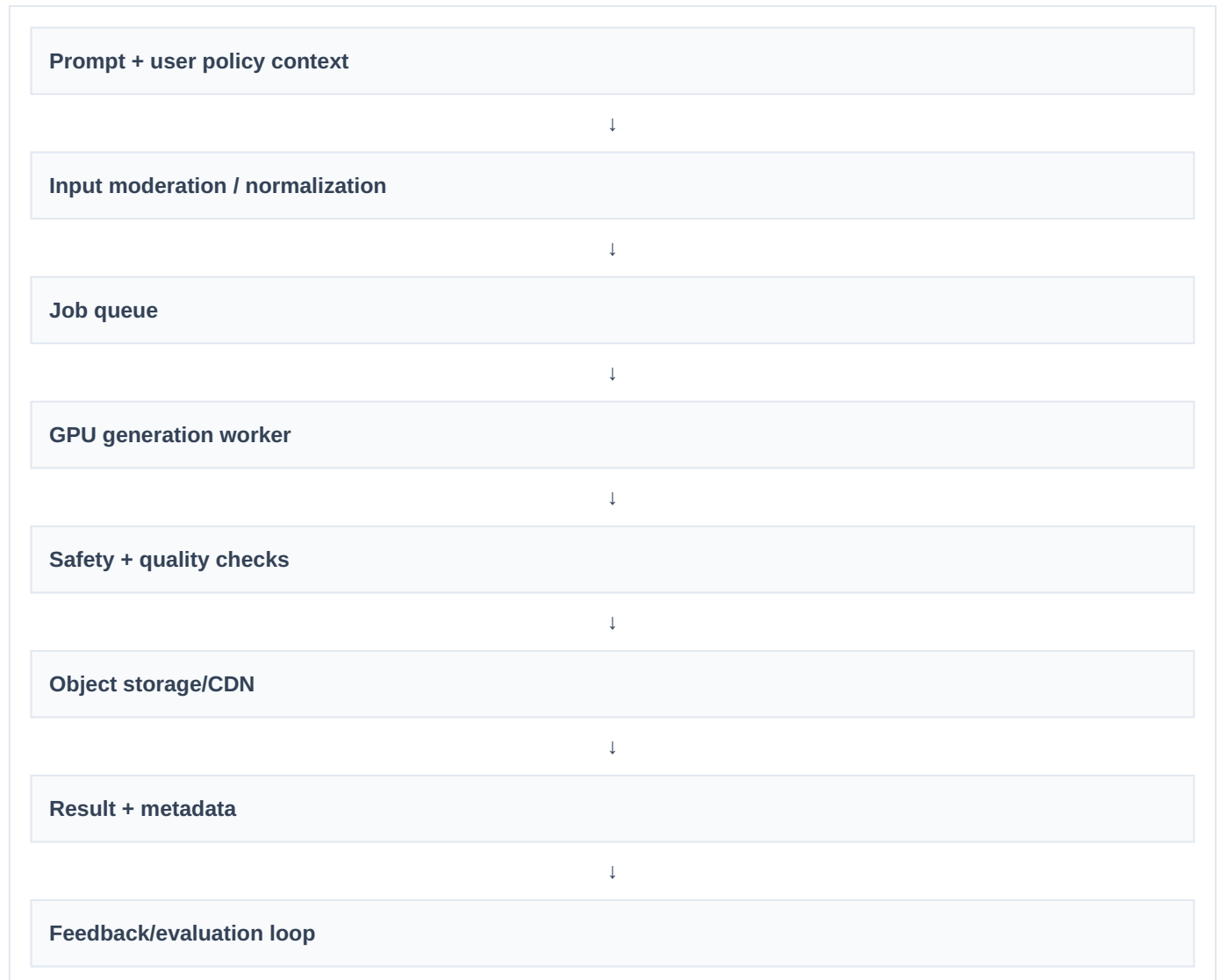
Generative systems can produce unsafe, copyrighted-looking or misleading content. Production layers may include input/output policy checks, provenance metadata, rate limits, abuse monitoring and human escalation.

A serving architecture also needs job queues for expensive generation, GPU scheduling, caching where safe, model/version tracking and reproducibility metadata such as seed and parameters.

Complete case walkthrough

Case study - production text-to-image platform

System: production text-to-image generation for a design application.



Discuss model/version selection, seed and parameter recording, asynchronous jobs, GPU utilization, prompt and output safety, provenance/watermarking policy, latency tiers, retries and cost controls. If users upload reference images, add privacy/retention rules and input scanning.

Additional interview drill

Answer each question aloud in 60-90 seconds. Then add one example, one failure mode and one trade-off.

#	Question	Strong answer should mention
1	VAE vs autoencoder?	probabilistic latent distribution and KL regularization
2	Why reparameterization?	differentiate through stochastic latent sampling
3	What is mode collapse?	GAN lacks diversity / covers few modes
4	Why latent diffusion?	lower-dimensional denoising for compute efficiency
5	What does CFG do?	strengthen conditioning, fidelity/diversity trade-off
6	ControlNet vs LoRA?	structural control vs parameter-efficient adaptation
7	How evaluate text-to-image?	human + automated + safety + diversity + latency/cost
8	Why are seeds not full reproducibility guarantees?	software/hardware/model changes and nondeterminism
9	How reduce unsafe output?	layered policy, model, filters, abuse monitoring, human review
10	How design async image generation?	request API, queue, GPU workers, object store, status/callback
11	When full fine-tune?	large high-quality data and strong domain shift with resources
12	What would you monitor?	failures, policy blocks, latency, GPU utilization, user quality, drift

Hands-on labs and worked exercises

These exercises turn the concepts into things you can calculate, code, debug or design. Work them before reading the expected reasoning.

Lab 1 - Sample a VAE latent variable

Goal Understand the reparameterization trick numerically.

Steps

1. Encoder outputs $\mu=0.5$ and $\sigma=0.2$.
2. Sample $\epsilon=-1.0$.
3. Compute $z=\mu+\sigma*\epsilon$.
4. Explain why gradients can flow to μ and σ .

Expected reasoning / result

$z=0.3$. Randomness lives in ϵ , while z is a differentiable function of the learned parameters.

Interview connection

This is the core reason VAEs can train with gradient descent despite sampling.

Lab 2 - Guidance trade-off experiment

Goal Reason about classifier-free guidance instead of treating it as a quality knob.

Steps

1. Generate the same prompt at low, medium and high guidance.
2. Score prompt adherence and diversity separately.
3. Record artifact rate.
4. Choose a default based on product objective.

Expected reasoning / result

Higher guidance can strengthen adherence while reducing diversity or causing artifacts. The optimal value is task/model dependent.

Interview connection

Name the trade-off explicitly.

Lab 3 - Choose adaptation method

GoalDistinguish prompting, ControlNet, LoRA and full fine-tuning.

Steps

1. Scenario A: enforce human pose from a skeleton.
2. Scenario B: teach a branded visual style.
3. Scenario C: change response behavior across many examples.
4. Pick the least expensive adequate method and explain why.

Expected reasoning / result

Pose → structural control such as ControlNet; style → LoRA can be appropriate; broad behavior/domain adaptation may justify larger fine-tuning depending on data and model.

Interview connection

Map method to the type of control required.

Lab 4 - Design an image-generation eval set

Goal Build multidimensional evaluation.

Steps

1. Create prompt slices: people, text rendering, spatial relations, rare concepts, safety-sensitive prompts.
2. Score prompt adherence, perceptual quality, diversity and policy compliance.
3. Include human pairwise preference for ambiguous quality.

Expected reasoning / result

No single automated metric captures all dimensions. A representative prompt suite plus human calibration is stronger.

Interview connection

Production GenAI evaluation is a portfolio of tests.

Lab 5 - Async generation service

Goal Design around expensive GPU work.

Steps

1. Accept request and validate policy.
2. Write a job to queue and return job ID.
3. GPU worker generates and stores artifact.
4. Run output checks.
5. Expose polling/callback status and immutable generation metadata.

Expected reasoning / result

Queues smooth bursts and decouple API latency from generation time. Retries need idempotent job IDs to avoid duplicate expensive generations.

Interview connection

Mention GPU scheduling, rate limits, provenance and failed-job recovery.

Last-minute revision sheet

Use this page in the final 30-60 minutes before an interview. Say each answer aloud in one or two sentences; if you cannot, revisit the relevant chapter.

Topic	One-line recall
VAE	Probabilistic latent encoder + reparameterization + decoder; reconstruction + KL regularization.
GAN	Generator vs discriminator game; sharp outputs but unstable training and possible mode collapse.
Diffusion	Train on noisy timesteps; sample by iterative reverse denoising, often in latent space.
CFG	Contrast conditional and unconditional denoising to strengthen prompt guidance; too high can hurt naturalness/diversity.
LoRA	Low-rank trainable adapters reduce trainable parameters and artifact size.
ControlNet	Adds spatial conditioning such as pose, depth or edges during denoising.
Evaluation	Measure fidelity, diversity, alignment, safety, latency and cost by representative slices.
Production	Queueing, GPU budgets, versioning, safety, provenance and observability matter as much as model choice.

Research basis and further reading

This handbook is original LearnMLAcademy material. Public interview reports and official technical documentation were used to identify current topic emphasis and production practices. The explanations, examples, diagrams and code in this handbook are independently written.

Hugging Face Diffusers - Training Overview

<https://huggingface.co/docs/diffusers/training/overview>

Current diffusion training patterns and supported workflows.

Hugging Face Diffusers - LoRA

<https://huggingface.co/docs/diffusers/en/training/lora>

Official parameter-efficient fine-tuning guidance for diffusion models.

Hugging Face Diffusers - ControlNet

<https://huggingface.co/docs/diffusers/api/pipelines/controlnet>

Official explanation of additional spatial conditioning for diffusion pipelines.

Aced / Exponent - Top Deep Learning Interview Questions (2026)

<https://www.tryexponent.com/blog/top-deep-learning-interview-questions>

Current deep-learning question bank reviewed by experienced ML engineers.

Google Machine Learning Crash Course

Official ML learning material emphasizes practical exercises, production systems, testing and monitoring in addition to model theory.

Made With ML MLOps Course

Production-oriented curriculum covering data, modeling, testing, reproducibility, serving, CI/CD and monitoring.

Current 2026 interview-preparation resources

Public guides increasingly emphasize production trade-offs, system design, debugging and role-specific reasoning rather than definition-only question banks.

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